# OUTCOMES

The Smart Elevator Crowd Management System successfully achieved the main objectives of the software prototype stage. The project demonstrates the integration of Artificial Intelligence, Computer Vision, local network communication and a real-time dashboard for intelligent lift request management.

- Successfully detected the presence of a person in the lift waiting area using YOLO-based person detection.
- Validated lift requests by checking whether a person remains present for a defined confirmation period.
- Prevented unnecessary lift movement when a button is pressed but no person is detected.
- Used multiple laptops as floor nodes and successfully communicated floor detection data through a common Wi-Fi network.
- Implemented a Flask-based central server to receive floor information, maintain pending requests and manage simulated lift movement.
- Developed a real-time dashboard displaying current floor, lift status, direction, target floor and floor-wise person detection status.
- Implemented W and S key simulation to represent UP and DOWN lift requests and tested continuous person-status updates.
- Successfully handled multiple pending floor requests and processed them sequentially based on the server decision logic.
- Provided a software foundation for future integration with Arduino/ESP32, motor control, limit switches and load-cell-based crowd detection.
- Demonstrated the proposed concept safely and economically using laptops and Wi-Fi without modifying a real elevator controller.

**Overall, the completed prototype proves the core intelligent decision-making logic of the smart elevator system, including person detection, request validation, multi-floor communication, server-side decision making and dashboard visualization. Physical lift movement and crowd-level detection are planned as the next development phase.**